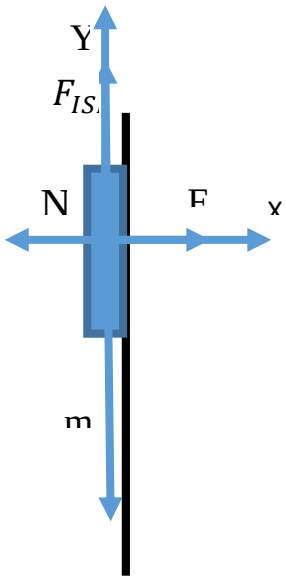
1. Bola kitobni vertikal devorga bosib turibdi. Devor va kitob orasidagi ishqalanish koeffitsiyenti 0,2 ga, kitobning massasi 2 kg ga teng. Kitob sirpanib ketmasligi uchun bola kitobni qanday minimal kuch bilan (N) bosib turishi kerak? $g = 10\ \frac{m}{s^2}$

Yechim:

Berilgan:



$\mu = 0{,}2$

$m = 2$ kg

$g = 10\ \frac{m}{s^2}$

F=?

Chizmaga ko'ra

Ikki o'q boyicha ikkita proyeksiya olamiz

X o'qi boyicha: $0 = F - N$

Y o'qi bo'yicha : $0 = mg - F_{ISH}$

Ishqalanish kuchi ifodas esa: $F_{ISH} = \mu N$

Birinchi tenglamadan: N=F

$F_{ISH} = \mu N = \mu F$

$mg = F_{ISH} = \mu F$

$F = \frac{mg}{\mu}$

$F = \frac{2\ kg * 10\frac{m}{s^2}}{0{,}2} = 100\ \ N$

Javob: $F = 100\ \ N$

2. 10 m balandlikdan tushayotgan 3 kg massali jismning yerga urilish vaqtidagi tezligi 12 m/s bo'lsa, u ishqalanish natijasida havoga necha J issiqlik miqdori ajratib chiqargan?

Yechim:

Berilgan :

h=10 m

m= 3kg

$v = 12\frac{m}{s}$

Q=?

Energiyaning saqlanish qonuniga ko'ra

$mgh = \frac{mv^2}{2} + Q$

$Q = mgh - \frac{mv^2}{2}$

$Q = 3\ kg * 10\frac{m}{s^2} * 10\ m - \frac{3\ kg}{2} * 144\frac{m^2}{s^2} = 84\ J$

Javob: $Q = 84\ J$

3. Mis va kumush qotishmasidan tayyorlangan buyumning havodagi og'irligi 2,5 N, suvdagi og'irligi 2,25 N. Qotishmaning o'rtacha zichligini (g/sm^3) toping.

Yechim:

Berilgan:

$mg = 2{,}5\ N$

$N = mg - F_A = 2{,}25\ N$

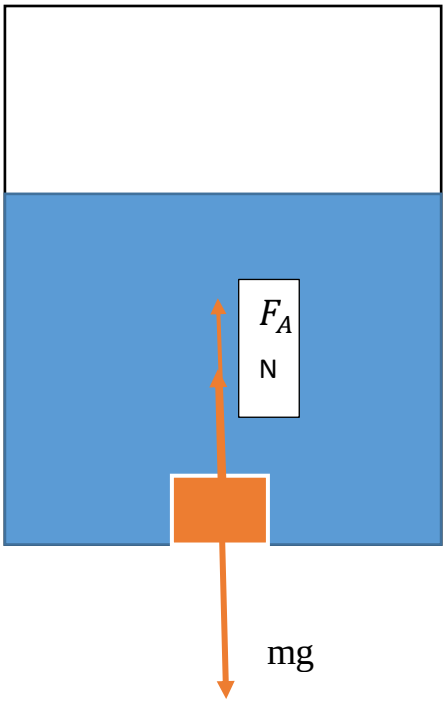
$\rho_j = ?$

$mg = \rho_j * V_j * g$

$F_A = \rho_s * V_j * g$

$N = \rho_j * V_j * g - \rho_s * V_j * g$

$N = V_j * g * (\rho_j - \rho_s)$



$$\frac{mg}{N} = \frac{\rho_j * V_j * g}{V_j * g * (\rho_j - \rho_s)} = \frac{\rho_j}{\rho_j - \rho_s}$$

$$\rho_j = \frac{mg * \rho_s}{mg - N}$$

$$\rho_j = \frac{mg * \rho_s}{mg - N} = 2{,}5\ N * \frac{1000\ \frac{kg}{m^3}}{0{,}25\ N} = 10000\frac{kg}{m^3}$$

Javob: $\rho_j = 10000\frac{kg}{m^3}$

4. Buyumning fokus masofasi 10 sm bo'lgan linzadan qanday masofaga joylashtirilsa, uning 5 marta kattalashgan tasviri hosil bo'ladi (sm)?

$$\frac{1}{F} = \frac{1}{d} \pm \frac{1}{f}$$

F=0,1 m

K=5 $\quad k = \frac{f}{d}$

d=? $\quad f = kd$

$$\frac{1}{F} = \frac{1}{d} \pm \frac{1}{kd} = \frac{k \pm 1}{kd}$$

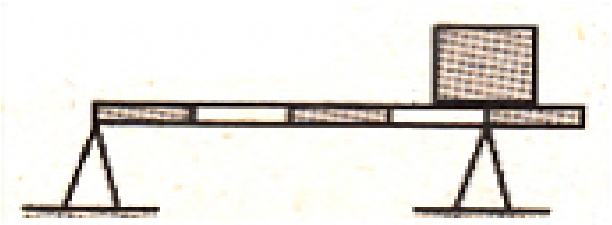
$$d = \frac{(k \pm 1)F}{k}$$

$$d_1 = 0{,}12\ m$$
$$d_2 = 0{,}8\ m$$

javob:

$$d_1 = 0{,}12\ m$$
$$d_2 = 0{,}8\ m$$

5. 40 kg massali balka ustiga 10 kg massali yuk qo'yilgan. O'ng tayanchda qanday reaksiya kuchi (N) hosil bo'ladi?



Yechim:

$m_b = 40\ kg$

$m_y = 10\ kg$

N=?

$$m_y g l_1 + m_b g l_2 = N l_1$$

$l_1 = 4l$

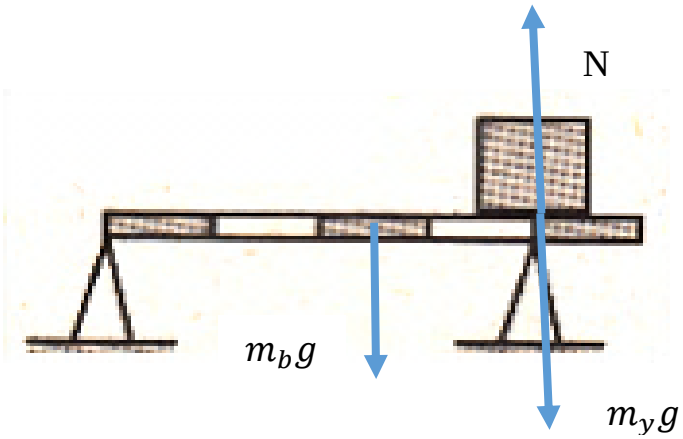
$l_2 = 2{,}5\ l$ bularni yuqoridagi tenglamaga qo'ysak.

$$m_y g 4l + m_b g 2{,}5\, l = 4lN$$

$$N = \frac{m_y g 4 + m_b g 2{,}5}{4} = \frac{1400\ N}{4} = 350\ N$$

Javob:

$N = 350N$



@olimpdep

Fan olimpiadalari bo'yicha
iqtidorli o'quvchilar bilan ishlash
DEPARTAMENTI